

Computer Programming, Grade 12

College Preparation

ICS4C

This course further develops students' computer programming skills. Students will learn object-oriented programming concepts, create object-oriented software solutions, and design graphical user interfaces. Student teams will plan and carry out a software development project using industry-standard programming tools and proper project management techniques. Students will also investigate ethical issues in computing and expand their understanding of environmental issues, emerging technologies, and computer-related careers.

Prerequisite: Introduction to Computer Programming, Grade 11, College Preparation

A. PROGRAMMING CONCEPTS AND SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** use data structures in the design and creation of computer programs;
- A2.** demonstrate the ability to use standard algorithms in the design and creation of computer programs;
- A3.** demonstrate an understanding of object-oriented programming concepts and practices in the design and creation of computer programs;
- A4.** create clear and accurate internal and external documentation to ensure the maintainability of computer software.

SPECIFIC EXPECTATIONS

A1. Data Structures

By the end of this course, students will:

- A1.1** perform operations on data types typically used in business applications (*e.g., express money amounts as dollars and cents, express dates and times in various formats*);
- A1.2** use Boolean operators (*e.g., AND, OR, NOT*), comparison operators (*i.e., equal to, not equal to, greater than, less than, greater than or equal to, less than or equal to*), arithmetic operators (*e.g., addition, subtraction, multiplication, division, exponentiation, parentheses*), and order of operations correctly in programming;
- A1.3** describe the structure of one-dimensional and two-dimensional arrays and related concepts including elements, indexes, and bounds.

A2. Using Standard Algorithms

By the end of this course, students will:

- A2.1** demonstrate the ability to manipulate and convert data in a computer program (*e.g., parse strings; convert data types such as numeric to string, and string to numeric; convert 'yes' or 'no' to Boolean*);
- A2.2** demonstrate the ability to read from, and write to, an external file (*e.g., sequential file, database, XML file, relational database via SQL*);

A2.3 demonstrate the ability to declare, initialize, modify, and access one-dimensional and two-dimensional arrays and elements within a program;

A2.4 demonstrate the ability to add, change, or delete elements of an array of objects in a program;

A2.5 demonstrate the ability to use search and sort routines (*e.g., `string.indexOf("cool")`, `Arrays.sort(intArray)`*) in a program.

A3. Object-oriented Programming

By the end of this course, students will:

- A3.1** explain the importance of designing reusable code in computer programs;
- A3.2** explain fundamental object-oriented programming concepts (*e.g., classes, objects, methods*);
- A3.3** apply the concepts of scope and visibility for variables, constants, and methods when creating classes in computer programs;
- A3.4** compare and contrast object-oriented and procedural programming paradigms.

A4. Code Maintenance

By the end of this course, students will:

- A4.1** write maintainable computer programs by creating clear and accurate internal documentation that provides in-depth explanations of complex blocks of code;
- A4.2** create clear and maintainable external user documentation (*e.g., Help file, how-to manual, FAQ, installation procedures, backup and recovery procedures, training materials*) as part of a complete software development project;
- A4.3** develop and implement a formal testing plan for a software development project to ensure program correctness.

B. SOFTWARE DEVELOPMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** design standard algorithms according to specifications;
- B2.** design software solutions using object-oriented programming concepts;
- B3.** design user-friendly graphical user interfaces (GUIs) that meet user requirements;
- B4.** participate in a large student-managed project, using proper project management tools and techniques to manage the process effectively.

SPECIFIC EXPECTATIONS

B1. Designing Standard Algorithms

By the end of this course, students will:

- B1.1** design algorithms to solve practical mathematical problems (*e.g., amount of paint or carpet needed, number of shingles needed, energy costs, amount of water needed for an aquarium, projection of Aboriginal youth population growth*);
- B1.2** design algorithms that require precision and accuracy when rounding numbers (*e.g., currency calculations, amortization, recipe volume changes*);
- B1.3** design data validation routines (*e.g., capitalization; formatting of postal codes, telephone numbers, SINS, and alphanumeric data; length and range checking*).

B2. Object-oriented Software Solutions

By the end of this course, students will:

- B2.1** demonstrate the ability to create and use instance methods (*e.g., constructors, mutators, accessors*) in a computer program;
- B2.2** design a simple base class to represent objects or concepts in a problem statement, using program templates or skeletons;
- B2.3** write methods that require parameter passing in a computer program.

B3. Graphical User Interfaces

By the end of this course, students will:

- B3.1** design graphical user interfaces that contain common controls (*e.g., buttons, labels, text boxes*);
- B3.2** design a user-friendly graphical user interface that helps to improve user accessibility (*e.g., for multilingualism; for those with limited eyesight or colour vision deficiency*);
- B3.3** evaluate a user interface for conformity with a given accessibility standard (*e.g., Section 508 of the Rehabilitation Act (US), W3C User Interface Domain, or a student- or teacher-created standard*);
- B3.4** design responses to user events in a graphical user interface.

B4. Student-managed Project

By the end of this course, students will:

- B4.1** describe the phases of a model (*e.g., waterfall, iterative, XP [Extreme Programming], RAD [Rapid Application Development]*) of the software development life cycle;
- B4.2** create a project plan for a software development project, outlining the tasks at each phase of the software development life cycle;
- B4.3** use project management tools (*e.g., Gantt chart, PERT chart*) and time management tools (*e.g., organizer, calendar*) to help develop a software project;
- B4.4** use industry-standard programming tools (*e.g., UML [Unified Modeling Language], diagrams, structure charts, flow charts, pseudocode*) to develop a software project.

C. PROGRAMMING ENVIRONMENT

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate the ability to use project management tools to plan and track activities for a software development project;
- C2.** demonstrate the ability to use software development tools to design and write a computer program.

SPECIFIC EXPECTATIONS

C1. Project Management Tools

By the end of this course, students will:

- C1.1** use software tools (*e.g., email, wikis, blogs, task lists, bulletin boards, spreadsheets, shared calendars*) to plan and track activities during a software development project;
- C1.2** communicate information about project status (*e.g., completed, in progress, not started, problems encountered, recommended modification to deadlines*) effectively in writing throughout the project.

C2. Software Development Tools

By the end of this course, students will:

- C2.1** use the features of a software development environment to debug programs and create functioning computer programs;
- C2.2** work independently, using the Help function, to resolve syntax issues while programming;
- C2.3** work independently, using reference materials (*e.g., code snippets, sample programs, APIs, tutorials*), to design and write functioning computer programs.

D. COMPUTERS AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** analyse and apply strategies that promote environmental stewardship with respect to the use of computers and related technologies;
- D2.** demonstrate an understanding of ethical issues and practices related to the use of computers;
- D3.** investigate and report on emerging computer technologies and their potential impact on society and the economy;
- D4.** research and report on the range of career paths and lifelong learning opportunities in software development or a computer-related field.

SPECIFIC EXPECTATIONS

D1. Environmental Stewardship and Sustainability

By the end of this course, students will:

- D1.1** outline and apply strategies to reduce the impact of computers and related technologies on the environment (*e.g., reduce, reuse, and recycle; turn computers and monitors off at end of day; participate in printer cartridge recycling*) and on human health (*e.g., ergonomic standards*);
- D1.2** investigate and describe governmental and community initiatives promoting environmental stewardship and sustainability (*e.g., local community recycling centres, private companies that refurbish computers, printer cartridge recycling programs*).

D2. Ethical Practices

By the end of this course, students will:

- D2.1** investigate and describe an ethical issue related to the use of computers (*e.g., piracy, privacy, security, phishing, spyware, cyberbullying*);
- D2.2** describe the essential elements of a code of ethics for computer programmers, and explain why there is a need for such a code (*e.g., plagiarism, backdoors, spyware, unethical programming practices*);
- D2.3** outline and apply strategies to encourage ethical computing practices at home, at school, and at work.

D3. Emerging Technologies

By the end of this course, students will:

- D3.1** describe the evolution of some emerging programming languages;
- D3.2** investigate and report on innovations in information technology (*e.g., webcasting, VoIP, multiplayer online gaming*) and their potential impact on society and the economy;
- D3.3** describe programming requirements for a variety of emerging technologies (*e.g., web programming, smartphones, embedded systems*).

D4. Computer-related Careers

By the end of this course, students will:

- D4.1** research and report on the range of career opportunities in software development, including duties, responsibilities, qualifications, and compensation;
- D4.2** research and report on opportunities for lifelong learning in software development or a computer-related field;
- D4.3** evaluate their own development of Essential Skills and work habits that are important for success in computer studies, as identified in the Ontario Skills Passport.